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Explainable Machine Leaning-based False Data Injection Classification Framework for AVs

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***Abstract*—With the increasing adoption of Autonomous Vehicles(AV) in modern transportation systems, there are risks associated with False Data injection (FDI) attacks, which target AV sensors.There is a need for an efficient system that can detect and identify this attack, ensuring the correct operations of AVs. We conducted experiments using a dataset containing FDI attack data for Industrial IoT (IIoT) environments integrated with AVs. This research proposes an effective approach for FDI attack detection and classification. Feature selection using SHapley Additive exPlanations (SHAP) was performed to increase the efficiency of ML models. To enhance the interpretability of the framework, we integrated Local Interpretable Model-Agnostic Explanations (LIME) based Explainable AI (XAI) techniques, providing transparency in attack detection. Our approach achieved a significant accuracy of 99% using the Random Forest classifier. The integration of XAI with ML model prediction helps in attaining high accuracy, explainability and efficient identification of FDI attacks. This study bridges the research gap by providing a solution that aids in detecting and neutralizing FDI attacks in AV systems and contributing to the security and reliability of autonomous operations.**

***Index Terms*—False Data Injection, Machine learning, Explainable AI, Autonomous Vehicles, SHAP, LIME**

I. Introduction

Transportation has come a long way, from horse-drawn carriages to modern cars, buses, and trucks. While these vehicles have made travel easier, they still depend on human drivers, who can make mistakes. In fact, human errors cause over 94% of road accidents [1]. Issues like tiredness, distraction, and reckless driving lead to crashes, injuries, and even deaths [2]. Bad driving also increases traffic, travel time, and fuel use, harming both the economy and the environment. To solve these problems, we need new technologies that can reduce human mistakes.

Autonomous vehicles (AVs) [3] are a major advancement in transportation, using artificial intelligence (AI), machine learning (ML), and advanced sensors to drive without human control. They rely on technologies like LiDAR (Light Detection and Ranging), radar, high-resolution cameras, ultrasonic sensors, and GPS (Global Positioning System) to constantly collect and analyze their surroundings. Smart ML models help these vehicles detect objects, predict the movement of other vehicles and pedestrians, and make quick driving decisions like braking, accelerating, and steering. AVs come in different levels, from basic driver assistance (such as cruise control and lane-keeping) to fully self-driving cars that need no human input. These technologies allow AVs to handle different road conditions, react faster than humans, and offer a safer and more efficient alternative to traditional vehicles [4].

AVs bring many benefits that solve the problems of human-driven transportation. One of the biggest advantages is safety, since AVs remove human errors, they can greatly reduce accidents and save lives [5]. They also improve traffic flow by choosing the best routes, which helps reduce congestion and travel time. AVs use fuel more efficiently, lowering pollution and supporting a cleaner environment by contributing to lower greenhouse gas emissions. Additionally, they offer better mobility for people with disabilities and the elderly, giving them more independence and accessibility. AVs could also change how cities are designed by reducing the need for large parking areas, making space for green areas and community spaces. As they become a key part of smart cities, AVs have the potential to create a more efficient and sustainable future for transportation [6].

Although AVs offer many benefits, they also face challenges that need to be solved for their safe and widespread use. One major issue is the accuracy of the data they rely on. AVs use real-time information from sensors, maps, and communication networks to make driving decisions. However, these systems can be targeted by cyber attacks, especially False Data Injection (FDI) attacks [7]. In an FDI attack, hackers manipulate sensor or network data, causing the vehicle to make dangerous mistakes like taking unsafe turns, misreading traffic signals, or failing to detect obstacles. These security risks raise concerns about the reliability of AVs, as incorrect data could put passengers and pedestrians in danger. Additionally, the high cost of AV technology, legal regulations, and ethical challenges in critical decision-making make their widespread adoption more complex [8].

AVs rely on vast amounts of sensor data from LiDAR, cameras, sensors, and GPS to navigate safely. However, this raw data is often noisy, inconsistent, and redundant, requiring preprocessing steps like cleaning, normalization, and feature extraction. This ensures that machine learning (ML) model can identify meaningful patterns for making real-time decisions in dynamic environments. While traditional ML approaches

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provide highly accurate predictions, they often function as black boxes, lacking transparency in their decision-making process. This lack of explainability is a major challenge, especially in critical situations like detecting FDI attacks, where trust and safety are essential. To overcome this, Explainable AI (XAI) techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) are integrated to make AV decision-making more understandable. XAI helps users trust AI decisions by providing explanations for how models reach their conclusions. XAI provides clear visualizations and explanations, making debugging easier and increasing trust among developers, regulators, and users. By combining high-performance reliability (PR) models with explainability, AV systems can achieve both accuracy and transparency, ensuring safer and more trustworthy operations in real-world conditions.

This study utilizes dataset derived from Industrial Internet of Things (IIoT) environments for False Data Injection (FDI) attack assessment in Autonomous Vehicles (AVs) due to their common technical characteristics. The networked communication method of IIoT systems resembles the way AVs use interconnected Iot sensing and camera devices. FDI attacks demonstrate equivalent patterns since they alter sensor information through equivalent procedures. Security for AV systems can benefit from the labeled attack data within the available dataset. The similar attack methods between these systems enable ML models to detect Foreign Deviated Inputs in AV systems using IIoT data samples. The dataset which was created for IIoT functions effectively with AV sensor vulnerabilities because of its matching feature structure. The study transforms the dataset into a tool suitable for AV-specific cybersecurity applications through SHAP feature selection along with ML classification methods.

Our proposed solution combines Machine Learning(ML) and Explainable AI (XAI) to detect and mitigate False Data Injection (FDI) attacks in Autonomous Vehicles (AVs) , providing a robust and interpretable security framework. Below is an overview of our approach:

- FDI attack detection in AV systems: Developed for detecting False Data Injection attacks targeting autonomous vehicle sensor data and control signals.
- Integration of ML models for Anomaly Detection: Implemented and compared multiple ML models to detect anomalies in AV sensor data.
- Explainability of ML model's with XAI: Addressed the "black box" nature of complex ML models by incorporating Explainable AI (XAI) techniques. These XAI methods were used to analyze the model's decisions and provide clear, interpretable explanations.

The rest of the paper is organised as follows: Section II explains the system model and problem formulation for the proposed approach. The explanation of the proposed approach is provided in Section III, while Section IV presents the performance evaluation of the algorithms used with respect to various parameters. At the end Section V concludes our research work.

II. System Model

The detailed system model for the proposed framework is presented in this section. Autonomous vehicle systems V utilizes IoT devices D for the functionality and security of components $D1$, $D2$, $D3$, $D4$ and upto Dn. There is a network attack for injecting false data A scenario targeting these components. The attack A comprises binary classification of benign data and network attacks for false data injection (FDI). a_1 denotes benign data, and a_2 denotes FDI attacks. The following equations represent the attack model:

$$D \in V, \tag{1}$$

$$D = (D1, D2, D3, D4, \ldots, Dn), \tag{2}$$

$$D \xleftarrow{\text{false data}} A, \tag{3}$$

$$A \in \{a_1, a_2\} \tag{4}$$

The goal is to increase the probability $\psi_{\max}$ of detection of the FDI attacks given IoT devices and the autonomous Vehicle. SHAP-based feature extraction increases the probability of predicting between the attack and no-attack data.

$$\psi_{\max} = P(A|D, V) \tag{5}$$

The data classified as benign ($D \notin A$) by the ML model is sent to the quantum layer Υ for a secure communication channel.

III. Proposed Approach

Fig. 1 represents the autonomous vehicle scenario, where external sources try to inject false data into IoT devices.Fig. 2 represents our proposed approach with and without XAI, feature extraction layer using SHAP, model predictions, and LIME explanation. Fig. 3 represents the complete system model with our proposed framework visually with a data collection layer, feature extraction using XAI layer and an ML layer.(Images and icons used are referenced from the online website [9].)

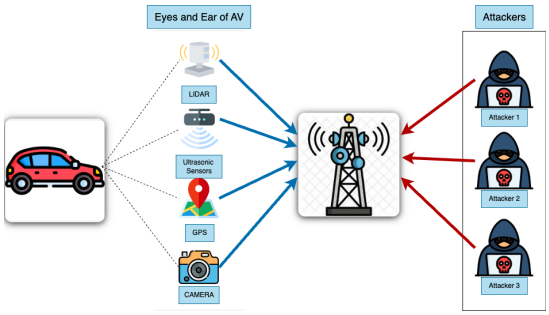

Fig. 1: FDI attack on AV.

A. Autonomous Vehicle System

In today's technology-driven world, AVs are rapidly transforming the transportation automation industry. AVs or self-driving cars rely heavily on IoT-enabled devices for

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real-time communication with centralized control systems for remote monitoring, ensuring efficiency and performance.Fig. 1 shows that our model includes components such as LIDAR, GPS, on-board cameras, ultrasonic sensors and vehicle-to-vehicle(V2V) communication modules, which work collaboratively to enable autonomous driving. These sensors act as the eyes and ears of the self-driving car as they gather continuous data about the environment. LIDAR (light detection and ranging) sensors provide 3D spatial mapping of environment by using laser pulses. However, LIDAR can be vulnerable to FDI attacks, where false and manipulated signals create artificial objects or can remove real objects which can lead to collisions. GPS (Global Positioning System) provides precise real-time location for navigation. Injecting false location data in GPS can mislead the AV to take incorrect routes. On-board cameras capture visual information about the road, traffic signs, pedestrians, and other vehicles. On FDI attacks in these camera sensors, it can deceive ML models into misclassification of objects. For example a stop sign can be misinterpreted as a speed limit sign due to FDI attack on these camera sensors. Ultrasonic Sensors help to detect nearby objects at close range. So, if false data is injected, it can cause collisions with nearby objects, for example, during parking. Vehicle-to-Vehicle(V2V) communication is a wireless technology that enables vehicles to share information such as speed and location with other vehicles, pedestrians, and infrastructure. It is a basic and essential technology for autonomous driving. There is risk of FDI attack on V2V communication modules, so if false messages are passed to AV there can be miscommunication between vehicles. For example, if traffic information is manipulated then the vehicle will make an incorrect decision and can create a hazard.

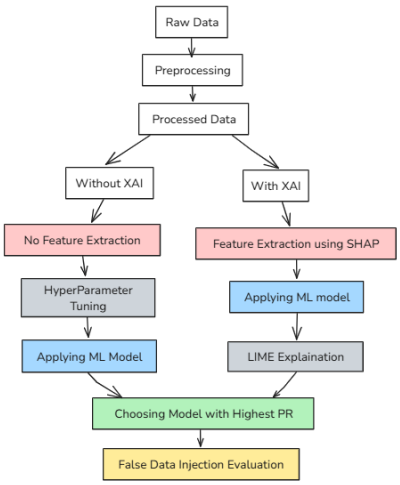

Fig. 2: XAI-based feature extraction and attack classification.

B. XAI based Feature Extraction and Attack Classification

1) Dataset Description: This section provides information about the dataset used, the preprocessing steps, and how the FDI attack classification takes place. The dataset used for training the model is an open dataset False Data Injection (FDI) attack for Industrial Internet of Things (IIoT) [10]. This dataset is a comprehensive resource that integrates data from diverse IoT devices, specifically focusing on FDI attacks. It was collected from a real-time IIoT infrastructure and includes both benign and FDI attack network behaviours. This data set offers a comprehensive description of actual scenarios. Initially, the dataset exhibited some inconsistencies, which were resolved through preprocessing using exploratory data analysis (EDA) techniques. The dataset comprises 15,425 rows, 26 features, and label class whether there is FDI attack or not.

2) Data Preprocessing: We removed specific attributes like ssl_issuer, ssl_subject etc., which contain only unique IDs. We performed null value analysis and numerical attributes analysis. The categorical features like markers are converted to numerical values. During training, 80% of the dataset is used, and 20% is used for inference time.

3) Comparing Traditional Approaches with XAI: Our methodology followed a two-phase approach to evaluate and enhance the performance of machine learning (ML) models. In the first phase, we trained six different ML models using the raw dataset and evaluated their performance in terms of accuracy and classification reports.

In the second phase, we applied eXplainable AI (XAI) techniques using SHAP (SHapley Additive exPlanations) to identify the top contributing features for the classification of FDI attacks. These top features were then used to retrain the same six ML models. The retrained models demonstrated improved accuracy and more detailed classification reports, showcasing the effectiveness of SHAP in feature extraction. The improvement can be attributed to SHAP's ability to identify the most impactful features, leading to better model optimization.

SHAP is a model-agnostic technique derived from cooperative game theory that explains the predictions made by complex ML models. It computes Shapley values, which quantify the marginal contribution of each feature to the model's output. This allows SHAP to provide local interpretability, helping to understand the reasons behind specific predictions. Additionally, SHAP visualizations make it easier to interpret the influence of individual features on model behavior.

By incorporating SHAP into the pipeline, we not only improved the performance of ML models but also enhanced transparency, confidence, and trust in the predictions by providing interpretable explanations for the models' decisions.

In Fig. 4, we illustrate the process of ranking features based on their impact on the output predictor. To quantify this impact, we employ SHAP values. Larger values correspond to higher importance for the associated attributes. The horizontal axis represents the transition of SHAP values from green to red. SHAP values measure the influence of each feature on the model's predictions. While the feature importance does not offer additional insights beyond the relative importance of attributes. Algorithm 1 explains the training and feature extraction part of our model.

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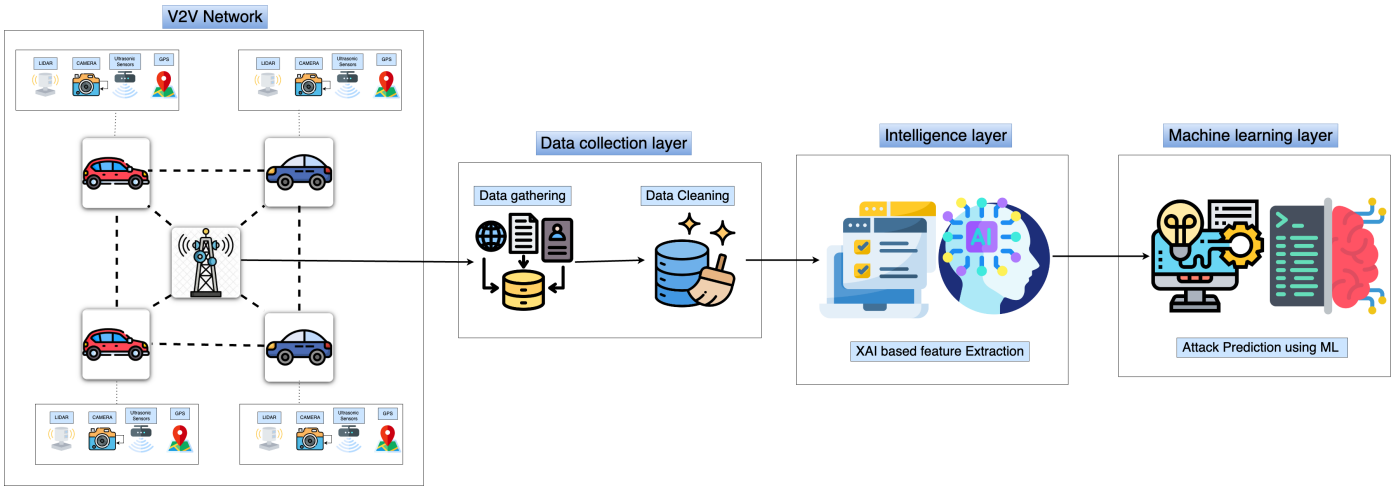

Fig. 3: AV System Model which describes our proposed framework

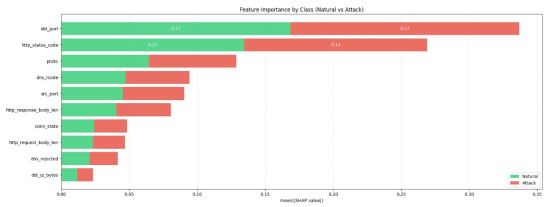

Fig. 4: Feature extraction using SHAP

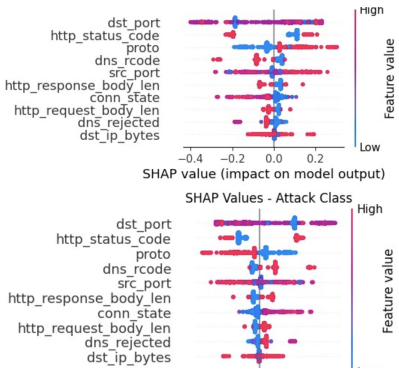

Fig. 5: SHAP normal vs attack comparison

Algorithm 1 Model Training and Feature Extraction

procedure TRAINANDEVALUATEMODELS(δ)
  $i \leftarrow 1$
  while $i \leq 6$ **do**
    $M_i \leftarrow$ TrainModel(δ)
    $Acc_i \leftarrow$ EvaluateModel(M_i, δ)
    $i \leftarrow i + 1$
  end while
  $F \leftarrow$ ExtractFeaturesUsingSHAP(δ)
  $i \leftarrow 1$
  while $i \leq 6$ **do**
    $M_i' \leftarrow$ RetrainModelWithFeatures(M_i, F)
    $Acc_i' \leftarrow$ EvaluateModel(M_i', F)
    $\Delta Acc_i \leftarrow Acc_i' - Acc_i$
    $i \leftarrow i + 1$
  end while
  return $M_i, Acc_i, F, \Delta Acc_i$
end procedure

By comparing feature importance between attack and regular network traffic in Fig. 5, the SHAP value plots show clear trends in how various network properties affect the model's predictions. Features such as destination port, protocol, and source port have more concentrated impact ranges between -0.2 and 0.4 in the attack class but show larger ranges of influence on model output in the benign class, with values ranging from -0.4 to 0.2. The difference between the distributions of SHAP values indicates that the model uses different feature patterns to differentiate between malicious and normal network behavior. In both cases, network protocol characteristics and port information are particularly important, but they show different patterns of importance.

ML Model Prediction:

$$\hat{a}_i = f(\mathbf{d}_i) \tag{6}$$

This equation predicts the probability distribution $\hat{a}_i$ over the label class for the ith instance using the ML model function $f(\mathbf{d})$, where $\mathbf{d}_i$ represents the feature vector of the ith instance.
SHAP Values Computation
Subset Generation:

$$\gamma \subseteq \mu \setminus \{i\} \tag{7}$$

This part generates subsets γ of features, excluding feature i, where γ ranges over all possible subsets of features except the ith feature.

Prediction Difference Calculation:

$$\Delta f_\gamma = f(\mathbf{d}_{S\cup\{i\}}) - f(\mathbf{d}_\gamma) \tag{8}$$

For each subset S, this part calculates the difference in model predictions (Δf_S) between two scenarios: one where feature i is included ($S \cup \{i\}$) and one where it is excluded (S).
Weighted Summation:

$$\Phi_i = \sum_\gamma \frac{|\gamma|!(\mu - |\gamma| - 1)!}{\mu!} \Delta f_\gamma \tag{9}$$

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The above equations are inspired from the original paper of SHAP [11]. The SHAP value Φ_i for the ith feature is computed as the weighted sum of all prediction differences (Δf_γ), where the weight is determined by the size of each subset S.

In the feature extraction process using SHAP, D represents the feature matrix of the dataset with dimensions $N \times \mu$, where N signifies the number of instances and μ denotes the number of features. Each feature vector $\mathbf{d}_i$ corresponds to the ith instance in the dataset, with dimensions $1 \times \mu$. a_i denotes the true label of the ith instance, indicating the attack type it belongs to, where a_i ranges from 1 to 2. The predicted probability distribution over the label class of benign or FDI attack for the ith instance is represented by $\hat{a}_i$, obtained from the ML model function $f(\mathbf{d})$. The function $\Phi(\mathbf{d}_i)$ computes the SHAP values for the ith instance using SHAP. Table I shows the difference between the traditional and XAI based approaches.

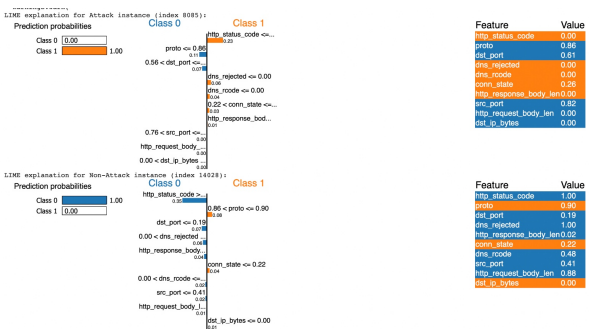

Fig. 6: Interpretability of ML model Using LIME

Local Interpretable Model-agnostic Explanations (LIME) is an approach that provides local, interpretable explanations for individual predictions by identifying the most influential features and helps to understand the decision-making process. In Fig. 6, we demonstrated the application of LIME for interpreting predictions in a binary classification task (FDI attack vs. non-attack). Attack Instance (Class 1) has confidence score of 1.0, features such as http_status_code, proto, and dst_port contributed significantly to the prediction (orange bars). For Non-Attack Instance (Class 0) with a confidence score of 1.0, features like http_status_code and proto were dominant contributors to the prediction. In this way, LIME helps researchers better understand and validate the decision-making process of ML models.

IV. Performance Evaluation

This section addresses the performance assessment of the suggested XAI-based method for smart home settings.

A. Simulation setup and tools

Our method uses Python v3.10.13 to simulate the system model on Kaggle Notebook. Data preprocessing is done with Pandas v2.1.1. Array operations are done with Numpy v1.26.4. Matplotlib v3.7.5 is used for data analysis and result visualization. The Apple Mac M2 Air, which has an 8-Core CPU, 8-Core GPU, and 8GB of RAM, is the machine utilized for simulation. Reliability is increased by maintaining the same setups during testing. Using their default parameters, we trained the ML models in TABLE I as well as our suggested model.

B. Performance analysis

The experimental examination of the system model was carried out by comparing the accuracy of several models in this part when features were retrieved with the aid of XAI employing SHAP. Tablename I displays the metrics used for performance analysis of various models. Prior to feature extraction using XAI, Random Forest's accuracy was 92.03%; this is greater than the 99.93% accuracy achieved by the Random Forest model. The results of our investigation into a number of additional evaluation metrics, including precision, recall, f1 score, and ROC-AUC curve, were consistent.In comparison to other models, this demonstrates how successful our suggested model is. Because of its strong predictive accuracy, robustness of the outliers, and versatility, Random Forest clearly outperformed all other models when we compared them. The Receiver Operating Characteristic

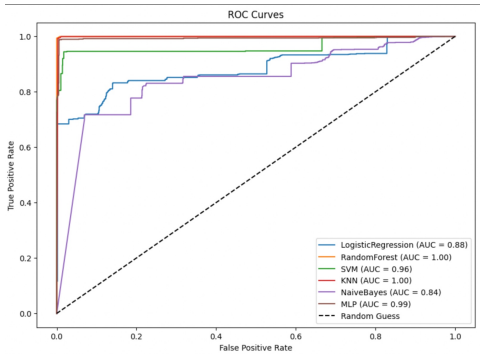

Fig. 7: ROC curve of different ML models used for FDI attack detection

TABLE I: Performance Metrics of Various Models

Model	Accuracy without XAI	Accuracy for XAI Features	Precis-ion	Recall	F1-Score
SVM	0.81	0.93	0.94	0.93	0.93
Logistic Regression (LR)	0.80	0.83	0.84	0.84	0.83
Naive Bayes	0.83	0.83	0.84	0.83	0.83
MLP Classifier	0.86	0.98	0.99	0.99	0.99
K-Nearest Neighbors (KNN)	0.91	0.99	0.99	0.99	0.99
Random Forest	**0.92**	**0.99**	**1.00**	**1.00**	**1.00**

(ROC) curve in Fig. 7 shows how well various classifiers work in our attack detection system. The primary performance statistic, the Area Under the Curve (AUC), is used to display the True Positive Rate (TPR) versus the False Positive Rate (FPR) at different thresholds. Random Forest has the best

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classification skill of all the models, with an AUC of 1.00. Random Forest (RF) constantly beats other models, such as K-Nearest Neighbors (KNN) and Multi-Layer Perceptron (MLP), despite their strong performance. Models with lower efficacy include Naïve Bayes and Logistic Regression. RF is the most dependable model for threat detection, as evidenced by its better AUC. The Precision-Recall (PR) curve in Fig. 8 assesses classifier performance, especially when there is imbalanced data. Recall assesses the model's capacity to identify positive cases, whereas precision calculates the percentage of correctly categorized positive instances. Random Forest's (RF) resilience in detection is shown by its consistent high precision across a range of recall values. While other models, such as MLP and KNN, also perform well, RF strikes the best consistent balance between recall and precision. Performance of Naïve Bayes and Logistic Regression noticeably declines, shows possible shortcomings in managing unbalanced datasets. The most dependable model for classification is further established by RF's high area under the PR curve.

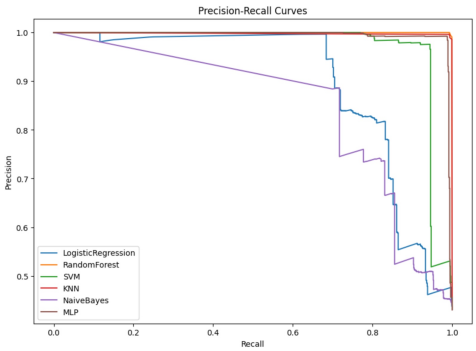

Fig. 8: Precision recall curves of different ML models for FDI attack detection

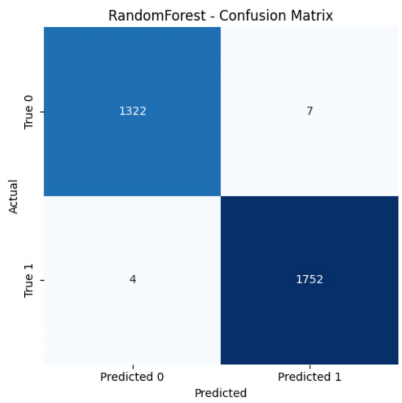

Fig. 9: Confusion Matrix of Random Forest

The confusion matrix of the Random Forest model's performance in categorizing the test data is shown in Fig. 9. With 1322 true negatives and 1752 true positives, the model exhibits a high degree of accuracy. Furthermore, the model's robustness and dependability are highlighted by the low frequency of misclassifications—7 false positives and 4 false negatives. This assessment demonstrates how well the model differentiates between the classes, which makes it a good fit for the specified classification task.

V. Conclusion

The paper proposes a unique method for classifying false data injection attacks in vehicular networks by combining machine learning algorithms with Explainable Artificial Intelligence (XAI) approaches. Our technology effectively adds a "brain" to the machine learning model by integrating XAI, which turns it from a black-box system into one that offers transparent, intelligible insights into its decision-making process. As demonstrated by extensive testing, this integration not only greatly improves detection accuracy and efficiency but also provides more transparency, overcoming the drawbacks of conventional techniques. This development ensures a more reliable and secure network by strengthening the security of vehicle communication. In the end, this study's XAI-driven feature extraction methods open the door to safer, more efficient, and comprehensible attack detection in vehicle networks.

In future, we intend to integrate a Federated Learning-based methodology to create a secure, privacy-preserving model for categorizing different kinds of attacks in vehicular communication networks and also adding blockchain for security. To further improve the system's usefulness, we also want to apply real-time attack classification in a real-world setting.

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